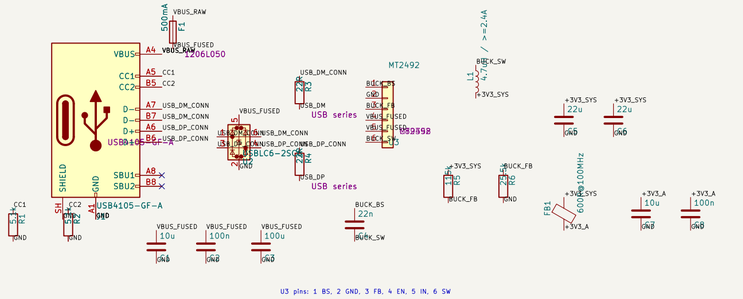
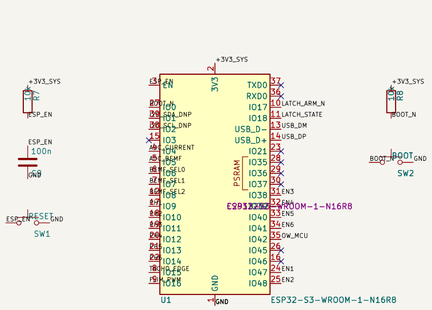
USB-C INPUT / 3V3 POWER

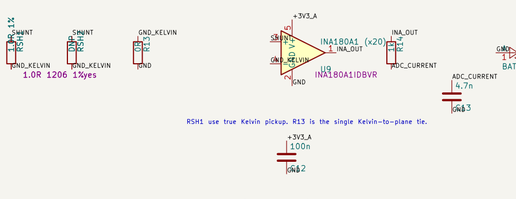


U3 pins: 1 BS, 2 GND, 3 FB, 4 EN, 5 IN, 6 SW

ESP32-S3 / USB / GPIO



SHARED SHUNT / CURRENT

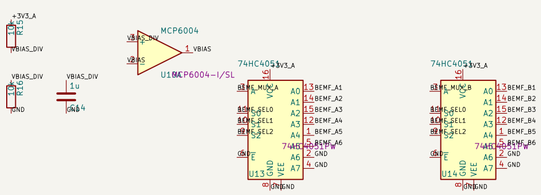


RSH1 use true Kelvin pickup. R13 is the single Kelvin-to-plane tie.

INA180 is 3V3-powered (intrinsic rail limit). R14 + D5 limit ADC injection transients.

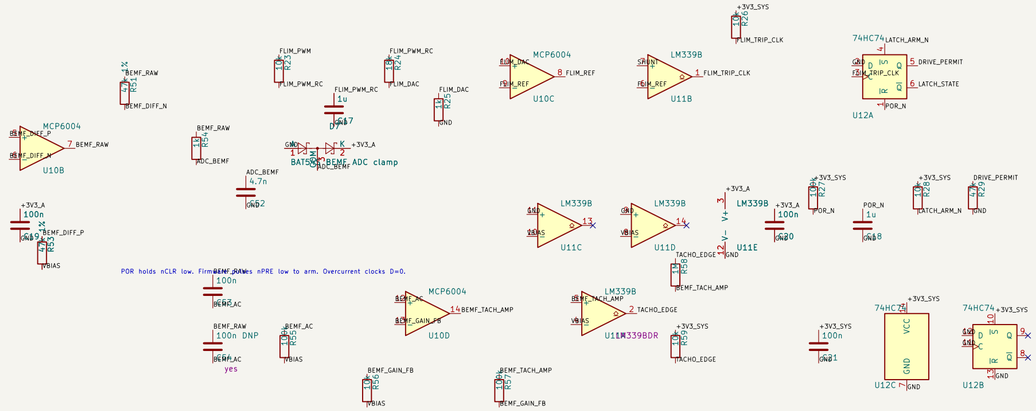
30mA=0.60V, 60mA=1.20V, 150mA=3.00V. Engagement current remains measurable.

SELECTED DIFFERENTIAL BEMF TACHO



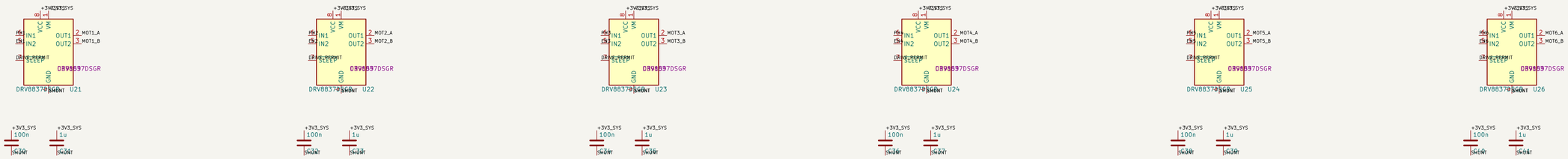
ADC_BEMF is differential motion evidence; TACHO_EDGE counts qualified commutation.

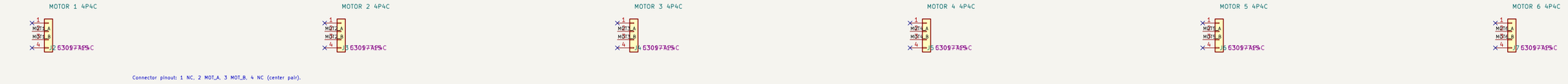
HARDWARE FORCE LIMIT / FAIL-SAFE LATCH



POR holds nCLR low. Firmware releases nPRE low to arm. Overcurrent clocks D=0.

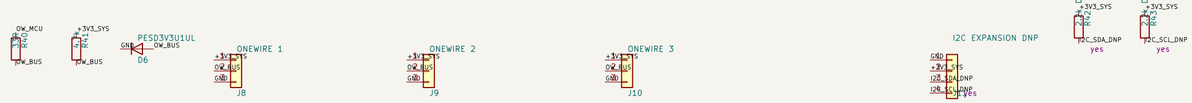
6 x DRV8837 / 4P4C MOTOR OUTPUTS





Connector pinout: 1 NC, 2 MOT_A, 3 MOT_B, 4 NC (center pair).

ONEWIRE / DNP I2C EXPANSION



OneWire pinout: 1 +3V3, 2 DQ, 3 GND. J11 pinout: GND, +3V3, SDA, SCL.

Sheet: /	
File: heatvalve-6-rev2.1.kicad_sch	
Title: HeatValve-6 Rev 2.1	
Size: A2	Date:
KiCad E.D.A. 10.0.3	**Rev:** Id: 1/1